

KENYATTA UNIVERSITY

SCHOOL OF ENGINEERING AND ARCHITECTURE

DEPARTMENT OF MECHANICAL ENGINEERING

CATS 1

Wednesday, 21/06/2023

EMM 312 FLUID MECHANICS III

Attempt the following QUESTION

QUESTION : (20 marks)

(a) Define the following terms with respect to open channel flow :-

(a) Shooting flow (2)

(b) Tranquil flow (2)

(b) Show that for a flow in an open rectangular channel, the critical depth (D_c) is given by
$$D_c = \frac{2}{3} H_c \quad \text{where } H_c = \text{critical specific Energy}$$

(6)

(c) An open water channel of triangular cross-section with a 90° included angle and a gradient of $\frac{1}{500}$ conveys water at the rate of ~~0.046535~~ $0.046535 \text{ m}^3/\text{sec}$. Determine the depth of flow in metres (10)

Assume the Chezy constant $C = 56$

Total = 20 marks

— 0 — 0 — 0 — 0 — 0 —

Write on both sides of the paper

Question.....

$$\cos 45 = \frac{D}{H} \Rightarrow H = \frac{D}{\cos 45} = \frac{D}{0.7071} = \frac{2D}{\sqrt{2}}$$

$$\tan 45 = \frac{x}{D} \Rightarrow x = D \tan 45 = D$$

$$A = \frac{1}{2} \times 2D \times D = D^2$$

$$P = \frac{R}{\cancel{0.7071}} \times 2 \times \frac{2D}{\sqrt{2}} \times 2$$

B

$$P = \frac{4D}{\sqrt{2}}$$

$$m = \frac{A}{P} = \frac{D^2}{\frac{4D}{\sqrt{2}}} = \frac{D^2 \sqrt{2}}{4D} = \frac{D \sqrt{2}}{4}$$

$$Q = A C \sqrt{m I}$$

$$0.046535 = D^2 \times 56 \sqrt{\frac{D \sqrt{2}}{4} \times \frac{1}{500}}$$

$$0.046535 = D^2 \times 56 \times 0.02659 \times \sqrt{D}$$

$$\frac{0.046535}{1.4891} = D^2 \sqrt{D}$$

$$0.03125 = D^2 \sqrt{D} = D^2 D^{\frac{1}{2}} = D^{\frac{5}{2}}$$

$$D = \sqrt[5]{0.03125}$$

$$D = 0.24999$$

$$D = \underline{\underline{0.25 \text{ m}}}$$

KENYATTA UNIVERSITY

SCHOOL OF ENGINEERING AND ARCHITECTURE

MECHANICAL ENGINEERING DEPARTMENT

CATS II

Thursday 6th July 2023.

EMM 312 FLUID MECHANICS III

Attempt the following question.

QUESTION: (20 marks)

(a) Show that in a frictionless adiabatic flow of a compressible fluid through a horizontal pipe, the Bernoulli's Equation is given by,

$$\left(\frac{\gamma}{\gamma-1}\right) \frac{p}{\rho} + \frac{V^2}{2} = \text{constant.}$$

where γ = adiabatic index of expansion / compression

p = pressure of the fluid

ρ = mass density of the fluid

V = velocity of the fluid.

10 marks

b) A horizontal air venturimeter is installed in a 150mm diameter pipe line whose throat diameter is 50mm. The inlet pressure is 965.7 kN/m² absolute and throat pressure is 896.7 kN/m² absolute.

The temperature at inlet was 15°C.

Calculate the mass of air flowing per second.

Assume $C_d = 1$

$$R = 287 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

10 marks

$$\gamma = 1.4$$

Total = 20 marks

$$\dot{m} = c_d a_1 p_1 \sqrt{\frac{2 \left(\frac{\gamma}{\gamma-1} \right) \frac{p_1}{\rho_1} (1 - \gamma_p^{\frac{\gamma-1}{\gamma}})}{\left(\frac{q_1}{q_2} \right)^2 \left(\frac{1}{\gamma_p} \right)^{\frac{2}{\gamma}} - 1}}$$

$$p = \rho R T$$

$$\gamma_p = \frac{p_2}{p_1}$$

$$\dot{m} = c_d a_1 \frac{p_1}{R T_1} \sqrt{\frac{2 \left(\frac{\gamma}{\gamma-1} \right) R T_1 (1 - \gamma_p^{\frac{\gamma-1}{\gamma}})}{\left(\frac{q_1}{q_2} \right)^2 \left(\frac{1}{\gamma_p} \right)^{\frac{2}{\gamma}} - 1}}$$

$$= 1 \left(\frac{\pi}{4} \times 0.15^2 \right) \left(\frac{965.7 \times 10^3}{287 \times 288} \right) \sqrt{\frac{2 \left(\frac{1.4}{0.4} \right) 287 \times 288 \left[1 - \left(\frac{896.7}{965.7} \right)^{\frac{0.4}{1.4}} \right]}{\left[\frac{\pi}{4} (0.15^2) \right]^2 \left[\frac{965.7}{896.7} \right]^{\frac{2}{1.4}} - 1}}$$

$$= 0.20646 \sqrt{\frac{12126.01646}{89.049}}$$

$$= 2.409 \text{ kg/s}$$

$$\therefore \dot{m} = 2.409 \text{ kg/s}$$

10

70